Exam

Name_____COMP1006: Engineering Programming_____

MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question.

1) The value of the expression '7' + '3' is the _____. 1) _____
A) string '10' B) integer 73. C) string '73' D) integer 10

2) What does the int function attempt to do in the following code? 2) _____

```
value = input('Enter an integer: ')
value = int(value)
```

A) Convert a non-object to an object. B) Convert an integer to a string.
C) Convert a string to an integer. D) None of the above.

3) Which of the following statements a), b) or c) is *false*? 3) _____
A) A condition is a Boolean expression with the value True or False.
B) Using a keyword as an identifier causes a ValueError.
C) True and False are keywords–words that Python reserves for its language features.
D) All of the above statements are *true*.

4) Which of the following statements is *true*? 4) _____
A) When you have many statements to execute as a group, you typically write them as a script stored in a file with the .py (short for Python) extension.
B) IPython interactive mode is helpful for executing brief code snippets and seeing immediate results.
C) The if statement uses a condition to decide whether to execute a statement (or a group of statements).
D) All of the above statements are *true*.

5) Which of the following statements is *false*? 5) _____
A) The *Style Guide for Python Code* states that each script should start with a docstring that explains the script's purpose.
B) If a line has a comment on it, that comment must begin the line with a hash character (#).
C) Comments do not cause the computer to perform any action when the code executes.
D) A line that begins with the hash character (#) is a comment.

6) Which of the following statements is *false*? 6) _____
A) Python ignores all white space.
B) You use blank lines and space characters to make code easier to read.
C) Together, blank lines, space characters and tab characters are known as white space.
D) Python allows you to split long code lines in parentheses without using continuation characters.

7) You may spread a lengthy statement over several lines with the _____ continuation character. 7) _____
A) ^ B) % C) @ D) \

8) Which of the following statements is *false*? 8) _____

A) Each suite contains zero or more statements.

B) The following if statement uses the == comparison operator to determine whether the values of variables number1 and number2 are equal:

```
if number1 == number2:
    print(number1, 'is equal to', number2)
```

C) Forgetting the colon (:) after the condition is a common syntax error.

D) Each if statement consists of the keyword if, the condition to test, and a colon (:) followed by an indented body called a suite.

9) Which of the following statements is *false*? 9) _____

A) Incorrect indentation of the statements in a suite can cause errors.

B) Using the assignment symbol (=) instead of the equality operator (==) in an if statement's condition is a common logic error.

C) Python requires you to indent the statements in suites.

D) Using == in place of = in an assignment statement can lead to subtle problems.

10) The chained comparison 3 < y <= 11 is *false* for which of the following values of y: 10) _____

A) 4 B) 10 C) 11 D) 3

SHORT ANSWER. Write the word or phrase that best completes each statement or answers the question.

11) There are no questions in this section. 11) _____

MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question.

12) Which of the following statements a), b) or c) is *false*? 12) _____

A) An algorithm is a procedure for solving a problem in terms of the actions to execute, and the order in which these actions execute.

B) You can solve any computing problem by executing a series of actions in a specific order.

C) Program control specifies the order in which statements (actions) execute in a program.

D) All of the above statements are *true*.

13) _____ is an informal English-like language for "thinking out" algorithms. 13) _____

A) Code B) Python C) Pseudocode D) Quasicode

14) Various Python statements enable you to specify that the next statement to execute may be other than the next one in sequence. This is called _____ and is achieved with Python control statements. 14) _____

A) transfer of control B) spread execution

C) alternate execution D) None of the above

15) The most important flowchart symbol is the _____, which indicates that a decision is to be made, such as in an if statement. 15) _____

A) flowline B) rectangle C) small circle D) diamond

16) Which of the following statements is *false*? 16) _____

A) The if...else statement performs an action if a condition is True or performs a different action if the condition is False.
B) Python provides three types of selection statements that execute code based on a condition–any expression that evaluates to either True or False.
C) Anywhere a single action can be placed, a group of actions can be placed.
D) The if…elif…else statement is called a double-selection statement because it selects one of two different actions (or groups of actions).

17) Python provides two iteration statements–_____ and _____. 17) _____

A) do while, for
B) do while, for each
C) while, for each
D) while, for

18) Which of the following statements a), b) or c) is *false*? 18) _____

A) You can construct any Python program from only six different forms of control (sequential execution, and the if, if…else, if…elif…else, while and for statements). You combine these in only two ways (control-statement stacking and control-statement nesting). This is the essence of simplicity.
B) You form each Python program by combining as many control statements of each type as you need for the algorithm the program implements.
C) With Single-entry/single-exit (one way in/one way out) control statements, the exit point of one connects to the entry point of the next. This is similar to the way a child stacks building blocks–hence, the term control-statement stacking.
D) All of the above statements are *true*.

19) Which of the following statements is *false*? 19) _____

A) Programs that are not uniformly indented are hard to read.
B) Indenting a suite is required; otherwise, an IndentationError syntax error occurs.
C) If you have more than one statement in a suite, those statements do not need to have the same indentation.
D) Sometimes error messages may not be clear. The fact that Python calls attention to the line is often enough for you to figure out what's wrong.

20) Which of the following statements is *false*? 20) _____

A) The diamond flowchart symbol has three flowlines emerging from it.
B) One flowline emerging from the diamond flowchart symbol indicates the direction to follow when the condition in the symbol is True. This points to the action (or group of actions) that should execute.
C) Another flowline emerging from the diamond flowchart symbol indicates the direction to follow when the condition is False. This skips the action (or group of actions).
D) The decision (diamond) symbol contains a condition that can be either True or False.

21) Which of the following statements is *false*? 21) _____

A) *Any* expression may be evaluated as True or False.
B) Strings containing characters are True and empty strings (", "" or """""") are False.
C) A condition which evaluates to a nonzero value is considered to be True, and a condition which evaluates to a value of zero is considered to be False.
D) All of the above statements are *true*.

22) Which of the following statements is *false*? 22) _____

A) You can replace if…else statements like the one above using a concise conditional expression:
result = ('Passed' if grade >= 60 else 'Failed')

B) The parentheses in a conditional expression are required; otherwise, a syntax error occurs.

C) Sometimes the suites in an if…else statement assign different values to a variable, based on a condition, as in:

```
grade = 87

if grade >= 60:
    result = 'Passed'
else:
    result = 'Failed'
```

D) All of the above statements are *true*.

23) Which of the following statements a), b) or c) is *false*? 23) _____

A) The following code shows two statements in the else suite of an if…else statement:

```
grade = 49

if grade >= 60:
    print('Passed')
else:
    print('Failed')
    print('You must take this course again')
```

B) In the code in Part (a), grade is less than 60, so *both* statements in the else's suite execute.

C) In the code of Part (a), even if you do not indent the else suite's second print statement, it's still in the else's suite. So, the code runs correctly.

D) All of the above statements are *true*.

24) Which of the following statements a), b) or c) is *false*? 24) _____

A) For a fatal logic error in a script, an exception occurs (such as a ZeroDivisionError from an attempt to divide by 0), so Python displays a traceback, then terminates the script.

B) A fatal error in interactive mode terminates only the current snippet. Then IPython waits for your next input.

C) With a logic error, an IPython interactive session terminates.

D) All of the above statements are *true*.

25) Which of the following statements is *false*? 25) _____

A) The following code sets product to the first power of 2 larger than 64:

```
product = 2

while product < 64:
    product = product * 2
```

B) Something in a while statement's suite must ensure that the condition eventually becomes False. Otherwise, a logic error called an infinite loop occurs.

C) The while statement allows you to *repeat* one or more actions while a condition remains True.

D) In applications executed from a Terminal, Command Prompt or shell, type *Ctrl+c* or *control + c* (depending on your keyboard) to terminate an infinite loop.

26) Which of the following statements is *false*? 26) _____

A) The end keyword argument is optional. If you do not include it, print uses a newline ('\n') by default.

B) Keyword arguments are sometimes called named arguments.

C) Built-in function print displays its argument(s), then moves the cursor to the next line. You can change this behavior with the argument end, as in

```
print(character, end=' ')
```

D) Python calls end a keyword argument, and end is a Python keyword.

27) What does the following line of code display? 27) _____

```
print(10, 20, 30, sep=', ')
```

A) 10,20,30 B) 102030 C) 10 20 30 D) 10, 20, 30

28) Which of the following statements is *false*? 28) _____

A) One of Python's most common iterable sequences is the list, which is a comma-separated collection of items enclosed in square brackets ([and]).

B) An iterable is an object from which the for statement can take one item at a time until no more items remain.

C) The sequence to the right of the for statement's keyword in must be an iterable.

D) The following code totals five integers in a list:

```
total = 0

for number in [2, -3, 0, 17, 9]:
    total + number
```

29) Which of the following statements is *false*? 29) _____

A) The iterator is like a bookmark–it always knows where it is in the sequence, so it can return the next item when it's called upon to do so.

B) The for statement uses the iterator "behind the scenes" to get each consecutive item until there are no more to process.

C) Each sequence has an iterator.

D) Lists are unordered and a list's items are mutable.

30) What does the following for statement print? 30) _____

```
for counter in range(10):
    print(counter, end=' ')
```

A) 0 1 2 3 4 5 6 7 8 9
B) 1 2 3 4 5 6 7 8 9
C) 0 1 2 3 4 5 6 7 8 9 10
D) It doesn't run because it's syntactically incorrect.

31) Which of the following statements a), b) or c) is *false*? 31) _____

A) Augmented assignments abbreviate assignment expressions in which the same variable name appears on the left and right of the assignment's =, as total does in:

```
for number in [1, 2, 3, 4, 5]:
    total = total + number
```

B) The statement f = f ** 3 may be replaced by the more concise augmented assignment statement
f *= 3.

C) The following code re-implements the for statement in Part (a) using an addition augmented assignment (+=) statement:

```
for number in [1, 2, 3, 4, 5]:
    total += number
```

D) All of the above statements are *true*.

32) Which of the following statements is *false*. 32) _____

A) Variables for totaling and counting normally are initialized to zero before they're used.
B) Once a correct algorithm has been specified, creating a working Python program from the algorithm is typically straightforward.
C) A requirements statement describes how a program should operate but not what the program is supposed to do.
D) Experience has shown that the most challenging part of solving a problem on a computer is developing an algorithm for the solution.

33) Which of the following statements is *false*? 33) _____

A) The following statement creates the variable grades and initializes it with a list of 10 integer grades.

```
grades = [98, 76, 71, 87, 83, 90, 57, 79, 82, 94]
```

B) Iteration is called *definite iteration* if the number of iterations is first known after the loop finishes executing.
C) Many scripts can be decomposed into initialization, processing and termination phases.
D) The *Style Guide for Python Code* recommends placing a blank line above and below each control statement.

34) Which of the following statements is *false*? 34) _____

A) The placeholder {average} converts the variable average's value to a string representation, then replaces {average} with that replacement text.

B) The following f-string inserts the value of average into a string:

```
f'Class average is {average}'
```

C) Replacement-text expressions in placeholders may contain values, variables or other expressions, such as calculations, but may not contain function calls.

D) You specify where to insert values in an f-string by using placeholders delimited by curly braces ({ and }).

35) Which of the following statements is *false*? 35) _____

A) Sentinel-controlled iteration is often called *indefinite* iteration because the number of iterations is *not* known before the loop begins executing.

B) Grades on a quiz are typically nonnegative integers between 0 and 100, so the value —1 is an acceptable sentinel value for grades data.

C) A sentinel value must match one acceptable input value.

D) A sentinel value indicates end of data entry."

36) Which of the following statements is *false*? 36) _____

A) The top specifies what should be done, but not how to implement it.

B) In the refinement process, we decompose the top into a sequence of smaller tasks–a process sometimes called divide and conquer.

C) In top-down, stepwise refinement, the top is a single statement that conveys the program's overall function but rarely conveys enough detail from which to write a program.

D) In the top-down, stepwise refinement process for developing algorithms, each refinement represents another portion of the algorithm.

37) Which of the following statements a), b) or c) is *false*? 37) _____

A) Formatting floating-point data with .2f *rounds* it to the tenths position.

B) The format specifier .2f formats data as a floating-point number (f) with two digits to the right of the decimal point (.2).

C) In an f-string, you can optionally follow a replacement-text expression with a colon (:) and a format specifier that describes how to format the replacement text.

D) All of the above statements are *true*.

38) With regard to the following code segment: 38) _____

```
# process 10 students
for student in range(10):
    # get one exam result
    result = int(input('Enter result (1=pass, 2=fail): '))

    if result == 1:
        passes = passes + 1
    else:
        failures = failures + 1
```

which of the following statements is *true*?

A) The if statement follows the for statement in sequence.
B) The for statement is nested in the if statement.
C) The if statement is nested in the for statement.
D) None of the above.

39) Which of the following statements is *false*? 39) _____

A) Function range's one-argument version produces a sequence of consecutive integers from 0 up to, but not including, the argument's value.
B) The following snippet produces the sequence 5 6 7 8 9.

```
for number in range(5, 10):
    print(number, end=' ')
```

C) The following snippet produces the sequence 10 8 6 4 2 0.

```
for number in range(10, 0, -2):
    print(number, end=' ')
```

D) The following snippet produces the sequence 0 2 4 6 8.

```
for number in range(0, 10, 2):
    print(number, end=' ')
```

40) Which of the following statements is *false*? 40) _____

A) Floating-point values are stored and represented precisely in binary format.
B) Financial institutions like banks that deal with millions or even billions of transactions per day have to tie out their transactions "to the penny." Floating-point numbers can represent some but not all monetary amounts with to-the-penny precision.
C) Many applications require precise representation of numbers with decimal points.
D) For monetary calculations and other applications that require precise representation and manipulation of numbers with decimal points, the Python Standard Library provides type Decimal, which uses a special coding scheme to solve the problem of "to-the-penny precision." Banks also have to deal with other issues such as using a fair rounding algorithm when they're calculating daily interest on accounts. Type Decimal offers such capabilities.

41) Which of the following statements a), b) or c) is *false*? 41) _____

A) The Python Standard Library is divided into modules–groups of related capabilities.

B) To use Decimal, you must first import its module, as in

```
import decimal
```

and refer to the Decimal type as decimal.Decimal, or you must indicate a specific capability to import using from…import, as in:

```
from decimal import Decimal
```

which imports only the type Decimal from the decimal module so that you can use it in your code.

C) The decimal module defines type Decimal and its capabilities.

D) All of the above statements are *true*.

42) Which of the following statements is *false*? 42) _____

A) Decimals support the standard arithmetic operators +, -, *, /, //, ** and %, as well as the corresponding augmented assignments.

B) You typically create a Decimal from a string.

C) You may perform arithmetic between Decimals and integers.

D) You may perform arithmetic between Decimals and floating-point numbers.

43) Which of the following statements is *false*? 43) _____

A) The following statement uses an f-string with two placeholders to format year and amount:

```
print(f'{year:>2}{amount:>10.2f}')
```

B) For single-digit year values 1—9, the format specifier >2 displays a value followed by the space character, thus right aligning the years in the first column.

C) The format specifier 10.2f in the placeholder {amount:>10.2f} formats amount as a floating-point number (f) right aligned (>) in a field width of 10 with a decimal point and two digits to the right of the decimal point (.2). Formatting a column of amounts this way aligns their decimal points vertically, as is typical with monetary amounts.

D) The placeholder {year:>2} uses the format specifier >2 to indicate that year's value should be right aligned (>) in a field of width 2–the field width specifies the number of character positions to use when displaying the value.

44) Which of the following statements is *false*? 44) _____

A) The following snippet produces the integer sequence 0 1 2 3 4 5 6 7 8 9:

```
for number in range(100):
    if number == 10:
        break
    print(number, end=' ')
```

B) The while and for statements each have an optional else clause that executes only if the loop terminates normally.

C) The following code snippet produces the sequence 0 1 2 3 4 5 5 6 7 8 9:

```
for number in range(10):
    if number == 5:
        continue
    print(number, end=' ')
```

D) Executing a break statement in a while or for immediately exits that statement.

45) Which of the following statements about the IPython session below is *true*? 45) _____

```
In [1]: gender = 'Female'

In [2]: age = 70

In [3]: if gender == 'Female' and age >= 65:
   ...:     print('Senior female')
   ...:
Senior female
```

A) The combined condition can be made clearer by adding redundant (unnecessary) parentheses

```
(gender == 'Female') and (age >= 65)
```

B) The right side of the and operator evaluates *only* if the left side is True.

C) The session defines two variables, then tests a condition that's True if and only if *both* simple conditions are True–if either (or both) of the simple conditions is False, the entire and expression is False.

D) All of the above statements are *true*.

46) Which of the following statements is *false*? 46) _____

A) Python stops evaluating an and-*expression* as soon as it knows whether the entire condition is False. Python stops evaluating an or-*expression* as soon as it knows whether the entire condition is True.

B) In operator expressions that use the and operator, make the condition that's more likely to be True the leftmost condition–in or operator expressions, make the condition that's more likely to be False the leftmost condition–each of these tactics can reduce a program's execution time.

C) The following condition stops evaluating immediately if gender is not equal to 'Female' because the entire expression must be False. If gender is equal to 'Female', execution continues, because the entire expression will be True if the age is greater than or equal to 65:

```
gender == 'Female' and age >= 65
```

D) The following condition stops evaluating immediately if semester_average is greater than or equal to 90 because the entire expression must be True:

```
semester_average >= 90 or final_exam >= 90
```

47) Which of the following statements is *false*? 47) _____

A) The if statement

```
if not grade == -1:
    print('The next grade is', grade)
```

also can be written as follows:

```
if grade != -1:
    print('The next grade is', grade)
```

B) You place the not operator before a condition to choose a path of execution if the original condition (without the not operator) is True.

C) The not operator is a unary operator–it has only one operand.

D) The Boolean not operator reverses the meaning of a condition–True becomes False and False becomes True.

48) Which of the following statements is *false*? 48) _____

A) The Python Standard Library's statistics module provides functions for calculating the mean, median and mode–these, too, are reductions.

B) The following session creates a list called grades, then uses the built-in sum and len functions to calculate the median "by hand"–sum calculates the total of the grades (397) and len returns the number of grades (5):

```
In [1]: grades = [85, 93, 45, 89, 85]

In [2]: sum(grades) / len(grades)
Out[2]: 79.4
```

C) The descriptive statistics mean, median and mode are measures of central tendency–each is a way of producing a single value that is in some sense typical of the others.

D) Like functions min and max, sum and len are both examples of functional-style programming reductions–they reduce a collection of values to a single value.

49) Which of the following statements is *false*? 49) _____

A) If a list's number of values is even, median returns the *mode* of the two middle values.

B) To help confirm the median and mode values of a grades list, you can use the built-in sorted function to get a copy of grades with its values arranged in increasing order, as in the following session, which makes it clear that both the median and the mode are 85:

```
In [1]: grades = [85, 93, 45, 89, 85]

In [2]: sorted(grades)
Out[2]: [45, 85, 85, 89, 93]
```

C) The argument of each of the statistics module's mean, median and mode functions must be an iterable.

D) The mode function causes a StatisticsError for lists like

```
[85, 93, 45, 89, 85, 93]
```

in which there are two or more "most frequent" values. Such a set of values is said to be bimodal.

50) Which of the following statements is *false*? 50) _____

A) Using existing functions as building blocks for creating new programs is a key aspect of software reusability–it's also a major benefit of object-oriented programming.

B) When you change a function's code, all calls to the function execute the updated version.

C) Packaging code as a function allows you to execute it from various locations in your program just by calling the function, rather than duplicating the possibly lengthy code.

D) Experience has shown that the best way to develop and maintain a large program is to construct it from a small number of large, proven pieces–this technique is called divide and conquer.

51) Which of the following statements is *false*? 51) _____

A) Each function should perform a single, well-defined task.

B) The following code calls function square twice. We've replaced each of the output values with ???. The first call produces the int value 49 and the second call produces the int value 6:

```
In [1]: def square(number):
    ...: """Calculate the square of number."""
    ...: return number ** 2
    ...:

In [2]: square(7)
Out[2]: ???

In [3]: square(2.5)
Out[3]: ???
```

C) Calling square with a non-numeric argument like 'hello' causes a TypeError because the exponentiation operator (**) works only with numeric values.

D) The statements defining a function are written only once, but may be called "to do their job" from many points in a program and as often as you like.

52) Which of the following statements is *false*? 52) _____

A) A function definition begins with the def keyword, followed by the function name, a set of parentheses and a colon (:).

B) The indented lines after the colon (:) are the function's suite, which consists of an optional docstring followed by the statements that perform the function's task.

C) Like variable identifiers, by convention function names should begin with a lowercase letter and in multiword names underscores should separate each word.

D) The required parentheses in a function definition contain the function's parameter list–a comma-separated list of parameters representing the data that the function needs to perform its task.

53) Which of the following statements is *false*? 53) _____

A) The following return statement first terminates the function, then squares number and gives the result back to the caller:

```
return number ** 2
```

B) Executing a return statement without an expression terminates the function and *implicitly* returns the value None to the caller. When there's no return statement in a function, it implicitly returns the value None after executing the last statement in the function's block.

C) Function calls can be embedded in expressions.

D) The following code calls square first, then print displays the result:

```
print('The square of 7 is', square(7))
```

54) Which of the following statements is *false*? 54) _____

A) Variables can be defined in a function's block.

B) Trying to access a local variable outside its function's block causes a NameError, indicating that the variable is not defined.

C) A function's parameters and variables are all local variables. They exist only while the function is executing and can be used only in that function.

D) All of the above statements are *true*.

55) Which of the following statements a), b) or c) is *false*? 55) _____

A) The call maximum(13.5, 'hello', 7) results in TypeError because strings and numbers cannot be compared to one another with the greater-than (>) operator.

B) The following session defines a maximum function that determines and returns the largest of three values–then calls the function three times with integers, floating-point numbers and strings, respectively.

```
In [1]: def maximum(value1, value2, value3):
   ...:     """Return the maximum of three values."""
   ...:     max_value = value1
   ...:     if value2 > max_value:
   ...:         max_value = value2
   ...:     if value3 > max_value:
   ...:         max_value = value3
   ...:     return max_value
   ...:

In [2]: maximum(12, 27, 36)
Out[2]: 36

In [3]: maximum(12.3, 45.6, 9.7)
Out[3]: 45.6

In [4]: maximum('yellow', 'red', 'orange')
Out[4]: 'yellow'
```

C) You also may call maximum with mixed types, such as ints and floats:

```
In [5]: maximum(13.5, -3, 7)
Out[5]: 13.5
```

D) All of the above statements are *true*.

56) Which of the following statements a), b) or c) is *false*? 56) _____

A) Using built-in functions or functions from the Python Standard Library's modules rather than writing your own can reduce development time and increase program reliability, portability and performance.

B) The built-in max and min functions know how to determine the largest and smallest of their two or more arguments, respectively.

C) The built-in max and min functions also can receive an iterable, such as a list but not a string.

D) All of the above statements are *true*.

57) Which of the following statements is *false*? 57) _____

A) Different values are likely to be displayed each time you run the code in Part (b).

B) The following code produces 10 random integers in the range 1—6 to simulate rolling a six-sided die:

```
import random

for roll in range(10):
    print(random.randrange(1, 7), end=' ')
```

C) You can introduce the element of chance via the Python Standard Library's random module.

D) Sometimes, you may want to guarantee reproducibility of a random sequence–for debugging, for example. You can do this with the random module's repeat function.

58) Which of the following statements a), b) or c) is *false*? 58) _____

A) The expression range(6,000,000) would be incorrect. Commas separate arguments in function calls, so Python would treat range(6,000,000) as a call to range with the *three* arguments 6, 0 and 0.

B) If randrange truly produces integers at random, every number in its range has an equal probability (or chance or likelihood) of being returned each time we call it.

C) We can use Python's underscore (_) digit separator to make a value like 6000000 more readable as 6_000_000.

D) All of the above statements are *true*.

59) Which of the following statements is *false*? 59) _____

A) In the following session, snippets [2] and [5] produce the same results purely by coincidence:

```
In [1]: import random

In [2]: random.seed(32)

In [3]: for roll in range(10):
   ...:     print(random.randrange(1, 7), end=' ')
   ...:
1 2 2 3 6 2 4 1 6 1
In [4] for roll in range(10):
   ...:     print(random.randrange(1, 7), end=' ')
   ...:
1 3 5 3 1 5 6 4 3 5
In [5] random.seed(32)

In [6]: for roll in range(10):
   ...:     print(random.randrange(1, 7), end=' ')
   ...:
1 2 2 3 6 2 4 1 6 1
```

B) You can use the random module's seed function to seed the random-number generator yourself–this forces randrange to begin calculating its pseudorandom number sequence from the seed you specify. Choosing the same seed will cause the random number generator to generate the same sequence of random numbers.

C) Function randrange actually generates pseudorandom numbers, based on an internal calculation that begins with a numeric value known as a seed.

D) When you're debugging logic errors in programs that use randomly generated data, it can be helpful to use the *same* sequence of random numbers until you've eliminated the logic errors, before testing the program with other values.

60) Which of the following statements is *false*? 60) _____

A) Like a list, a tuple is a sequence.
B) To avoid name collisions, variables in different functions' blocks must have different names.
C) To use a tuple's values, you can assign them to a comma-separated list of variables, which unpacks the tuple.
D) Each local variable is accessible only in the block that defined it.

61) Which of the following statements is *false*? 61) _____

A) The in operator in the following expression tests whether the tuple (7,11) contains sum_of_dice's value. The operator's right operand can be any iterable:

```
sum_of_dice in (7, 11)
```

B) The concise condition in Part (a) is equivalent to

```
(sum_of_dice = 7) or (sum_of_dice = 11)
```

C) There's also a not in operator to determine whether a value is *not* in an iterable.
D) All of the above statements are *true*.

62) Which of the following statements is *false*? 62) _____

A) A module is a file that groups related functions, data and classes.
B) A package groups related modules. Every Python source-code (.py) file you create is a module. They're typically used to organize a large library's functionality into smaller subsets that are easier to maintain and can be imported separately for convenience.
C) A key programming goal is to avoid "reinventing the wheel."
D) The Python Standard Library module money provides arithmetic capabilities for performing monetary calculations.

63) Which of the following statements is *false*? 63) _____

A) The following snippet calculates the absolute value of -10 by calling the math module's fabs function, which returns its result as a float value:

```
math.fabs(-10)
```

B) The value of the expression floor(—3.14159) is —3.0.
C) The following snippet calculates the square root of 900 by calling the math module's sqrt function, which returns its result as a float value:

```
math.sqrt(900)
```

D) An import statement of the following form enables you to use a module's definitions via the module's name and a dot (.):

```
import math
```

64) Which of the following statements a), b) or c) is *false*? 64) _____

A) IPython tab completion results may vary based on your operating system platform and what you have imported into your IPython session.
B) You can view a module's documentation in IPython interactive mode via tab completion–a discovery feature that speeds your coding and learning processes.
C) After you type a portion of an identifier and press *Tab*, IPython completes the identifier for you or provides a list of identifiers that begin with what you've typed so far.
D) All of the above statements are *true*.

65) Which of the following statements is *false*? 65) _____

A) In the math module, pi and e represent the mathematical constants π and *e*, respectively.
B) Python does not have constants, so even though pi and e are real-world constants, you must not assign new values to them, because that would change their values.
C) To help distinguish "constants" from other variables, the Python style guide recommends naming your custom constants with a leading underscore (_) and a trailing underscore.
D) To view a list of identifiers defined in a module, type the module's name and a dot (.), then press *Tab*.

66) Which of the following statements is *false*? 66) _____

A) When defining a function, you can specify that a parameter has a default parameter value.
B) The call rectangle_area() to the function in Part C) returns the value 0 (zero).
C) The following defines a function rectangle_area with default parameter values:

```
def rectangle_area(length=2, width=3):
    """Return a rectangle's area."""
    return length * width
```

D) When calling the function, if you omit the argument for a parameter with a default parameter value, the default value for that parameter is automatically passed.

67) Assuming the following function definition, which of the following statements is *false*? 67) _____

```
def rectangle_area(length=2, width=3):
    """Return a rectangle's area."""
    return length * width
```

A) Any parameters with default parameter values must appear in the parameter list to the *right* of parameters that do not have defaults.
B) For the following call, the interpreter passes the default parameter value 3 for the width as if you had called rectangle_area(3, 10):

```
rectangle_area(10)
```

C) You specify a default parameter value by following a parameter's name with an = and a value.
D) The following call to rectangle_area has arguments for both length and width, so IPython ignores the default parameter values:

```
rectangle_area(10, 5)
```

68) Based on the following function definition: 68) _____

```
def rectangle_area(length, width):
    """Return a rectangle's area."""
    return length * width
```

Which of the following statements based on this function definition is *false*?

A) In each function call, you must place keyword arguments *after* a function's positional arguments–that is, any arguments for which you do not specify the parameter name. Such arguments are assigned to the function's parameters left-to-right, based on the argument's positions in the argument list.

B) Keyword arguments can improve the readability of function calls, especially for functions with many arguments.

C) The following call shows that the order of keyword arguments matters–they need to match the corresponding parameters' positions in the function definition:

```
rectangle_area(width=5, length=10)
```

D) Each keyword *argument in a call* has the form *parametername=value*.

69) Which of the following statements a), b) or c) is *false*? 69) _____

A) The min function's documentation states that min has two *required* parameters (named arg1 and arg2) and an optional third parameter of the form *args, indicating that the function can receive any number of additional arguments.

B) The following is a valid call to function min

```
min(88)
```

C) The * before the parameter name *args in Part (a) tells Python to pack any remaining arguments into a tuple that's passed to the args parameter.

D) All of the above statements are *true*.

70) Based on the following function definition that can receive an arbitrary number of arguments: 70) _____

```
In [1]: def average(*args):
   ...:     return sum(args) / len(args)
   ...:
```

Which of the following statements a), b) or c) is *false*?

A) The following session calls average several times confirming that it works with arbitrary argument lists of different lengths:

```
In [2]: average(5, 10)
Out[2]: 7.5

In [3]: average(5, 10, 15)
Out[3]: 10.0

In [4]: average(5, 10, 15, 20)
Out[4]: 12.5
```

B) If the function has multiple parameters, the *args parameter must be the *leftmost* one.

C) The parameter name args is used by convention, but you may use any identifier.

D) All of the above statements are *true*.

71) Which of the following statements a), b) or c) is *false*? 71) _____

A) A method is simply a function that you call on an object using the form

object_name.method_name(*arguments*)

B) After the preceding session, s contains 'HELLO'.

C) The following session calls the string object s's lower and upper methods, which produce *new* strings containing all-lowercase and all-uppercase versions of the original string, leaving s unchanged:

```
In [1]: s = 'Hello'

In [2]: s.lower() # call lower method on string object s
Out[2]: 'hello'

In [3]: s.upper()
Out[3]: 'HELLO'
```

D) All of the above statements are *true*.

72) Which of the following statements a), b) or c) is *false*? 72) _____

A) A local variable's identifier has local scope. It's "in scope" only from its definition to the end of the program. It "goes out of scope" when the function returns to its caller.

B) A local variable can be used only inside the function that defines it.

C) Each identifier has a scope that determines where you can use it in your program. For that portion of the program, the identifier is said to be "in scope."

D) All of the above statements are *true*.

73) Which of the following statements a), b) or c) is *false*? 73) _____

A) Identifiers with global scope can be used in a .py file or interactive session anywhere after they're defined.

B) Variables with global scope are known as global variables.

C) Identifiers defined outside any function (or class) have *script* scope–these may include functions, variables and classes.

D) All of the above statements are *true*.

74) Which of the following statements a), b) or c) is *false*? 74) _____

A) To modify a global variable in a function's block, you must use a global statement to declare that the variable is defined in the global scope.

B) You can access a global variable's value inside a function.

C) By default, you cannot *modify* a global variable in a function–when you first assign a value to a variable in a function's block, Python creates a *new* local variable.

D) All of the above statements are *true*.

75) Which of the following statements is *false*? 75) _____

A) Importing a module's identifiers with a wildcard import can lead to subtle errors–it's considered a dangerous practice that you should avoid.

B) You can import *all* identifiers defined in a module with a wildcard import of the form from *modulename* import *

C) The following session is a safe use of a wildcard import:

```
In [1]: e = 'hello'

In [2]: from math import *

In [3]: e
Out[3]: 2.718281828459045
```

D) A wildcard import makes all of the module's identifiers available for use in your code.

76) Which of the following statements is *false*? 76) _____

A) import…as is frequently used to import Python libraries with convenient abbreviations, like stats for the statistics module.

B) The numpy module is typically imported with

```
import numpy as npy
```

C) Sometimes it's helpful to import a module and use an abbreviation for it to simplify your code. The import statement's as clause allows you to specify the name used to reference the module's identifiers. For example, we can import the statistics module and access its mean function as follows:

```
In [1]: import statistics as stats

In [2]: grades = [85, 93, 45, 87, 93]

In [3]: stats.mean(grades)
Out[3]: 80.6
```

D) Typically, when importing a module, you should use import or import…as statements, then access the module through the module name or the abbreviation following the as keyword, respectively. This ensures that you do not accidentally import an identifier that conflicts with one in your code.

77) Which of the following statements is *false*? 77) _____

A) With pass-by-value, the called function receives a *copy* of the argument's *value* and works exclusively with that copy. Changes to the called function's copy do *not* affect the original variable's value in the caller.

B) When a function call provides an argument, Python copies a *reference* to the argument object–not the object itself–into the corresponding parameter–functions often manipulate large objects–frequently copying them would consume large amounts of computer memory and significantly slow program performance.

C) Python arguments are always passed by value.

D) With pass-by-reference, the called function can access the argument's value in the caller directly and modify the value if it's mutable.

78) Which of the following statements is *false*? 78) _____

A) Two objects can reside at the same memory address.
B) No two objects in memory can have the same *identity*.
C) The integer result of calling id is known as the object's identity.
D) Though we can't see an object's address, we can use the built-in id function to obtain a *unique* int value which identifies only that object while it remains in memory.

79) Which of the following statements a), b) or c) is *false?* 79) _____

A) The top stack frame always contains the information the currently executing function needs to return control to its caller.
B) For each function call, the interpreter pushes a stack frame onto the stack. This entry contains the return location that the called function needs so it can return control to its caller. When the function finishes executing, the interpreter pops the function's stack frame, and control transfers to the return location that was popped.
C) If before a function returns it makes a call to another function, the interpreter pushes a stack frame for that function call onto the stack. Thus, the return address required by the newly called function to return to its caller is now on top of the stack.
D) All of the above statements are *true*.

80) Which of the following statements is *false*? 80) _____

A) A called function's stack frame is popped when the function is called and exists while the function is executing.
B) A called function's stack frame is the perfect place to reserve memory for the function's local variables.
C) When a function returns, it no longer needs its local variables, so its stack frame is *popped* from the stack, and its local variables no longer exist.
D) Most functions have one or more *parameters* and possibly local variables that need to exist while the function is executing, remain active if the function makes calls to other functions, and "go away" when the function returns to its caller.

81) Which of the following statements a), b) or c) is *false*? 81) _____

A) You've already used list, string and built-in function range iterators with the for statement, and you've used iterators with several reductions (functions sum, len, min and max).
B) Functional-style programs can be easier to parallelize to get better performance on today's multi-core processors.
C) Like other popular languages, such as Java and C#, Python is a purely functional language.
D) All of the above statements are *true*.

82) Which of the following statements about functional-style programming is *false*? 82) _____

A) Typically, library code handles the *how* for you. This can eliminate many errors.
B) Functional-style programming emphasizes mutability–it uses only operations that modify variables' values.
C) It lets you simply say *what* you want to do. It hides many details of *how* to perform each task.
D) Consider the for statement in many other programming languages. Typically, *you* must specify all the details of counter-controlled iteration: a control variable, its initial value, how to increment it and a loop-continuation condition that uses the control variable to determine whether to continue iterating. This style of iteration is known as external iteration and is error-prone.

83) Which of the following statements is *false*? 83) _____

A) Pure functions can have side effects–for example, if you pass a mutable list to a pure function, the list can contain different values before and after the function call.

B) In pure functional programming languages you focus on writing pure functions. A pure function's result depends only on the argument(s) you pass to it. Also, given particular arguments, a pure function always produces the same result. For example, built-in function sum's return value depends only on the iterable you pass to it.

C) Functions are objects that you can pass to other functions as data.

D) The following session demonstrates that when you call the pure function sum, it does not modify its argument.

```
In [1]: values = [1, 2, 3]

In [2]: sum(values)
Out[2]: 6

In [3]: sum(values) # same call always returns same result
Out[3]: 6

In [4]: values
Out[4]: [1, 2, 3]
```

84) Which of the following statements is *false*? 84) _____

A) For practical reasons, polling organizations trying to predict who will become the next president work with carefully selected small subsets of the population known as samples. Many of the polls in the 2016 election had sample sizes of about 1000 people.

B) Sometimes a population is quite large, such as the people likely to vote in the next U.S. presidential election–in excess of 100,000,000 people.

C) Measures of dispersion help you understand how concentrated the values are.

D) When we're talking about a group, the entire group is called the population.

85) Which of the following statements a), b) or c) is *false*? 85) _____

A) The smaller the variance and standard deviation are, the further the data values are from the mean and the greater overall dispersion (that is, spread) there is between the values and the mean.

B) The standard deviation is the square root of the variance, which tones down the effect of the outliers.

C) The following code calculates the population standard deviation with the statistics module's pstdev function:

```
statistics.pstdev([1, 3, 4, 2, 6, 5, 3, 4, 5, 2])
```

D) All of the above statements are *true*.

86) Which of the following statements is *false*? 86) _____

A) Examples of collections include your favorite songs on your smartphone, your contacts list, a library's books, your cards in a card game, your favorite sports team's players, the stocks in an investment portfolio, patients in a cancer study and a shopping list.

B) Collections are prepackaged data structures consisting of related data items.

C) Lists are modifiable and tuples are not. Each can hold items of the same or different types.

D) Tuples can dynamically resize as necessary, growing and shrinking at execution time.

87) Lists may store _____ data, that is, data of many different types. 87) _____

A) heterogeneous B) parallel
C) homogeneous D) None of the above.

88) Consider the list c: 88) _____

```
c = [-45, 6, 0, 72, 1543]
```

Which of the following statements a), b) or c) is *false*?

A) The length of c is 5.
B) You reference a list element by writing the list's name followed by the element's index (that is, its position number) enclosed in square brackets.
C) The names of c's elements are c[0], c[1], c[2], c[3] and c[4].
D) All of the above statements are *true*.

89) Which of the following statements is *false*? 89) _____

A) You can insert and delete list elements, changing the list's length.
B) Python's string and tuple sequences are immutable–they cannot be modified.
C) Lists are.
D) You can get the individual characters in a string, and you can assign a new value to one of the string's characters.

90) Which of the following statements a), b) or c) is *false*? 90) _____

A) You can concatenate two lists, two tuples or two strings using the + operator. The result is a *new* sequence of the same type containing the left operand's elements followed by the right operand's elements.
B) List elements can be accessed via their indices and the subscription operator ([]).
C) A TypeError occurs if the + operator's operands are different sequence types–for example, concatenating a list and a tuple is an error.
D) All of the above statements are *true*.

91) We've replaced the results of the four list comparisons in snippets [4] through [7] below with ???. What are those four values? 91) _____

```
In [1]: a = [1, 2, 3]

In [2]: b = [1, 2, 3]

In [3]: c = [1, 2, 3, 4]

In [4]: a == b
Out[4]: ???

In [5]: a == c
Out[5]: ???

In [6]: a < c
Out[6]: ???

In [7]: c >= b
Out[7]: ???
```

A) True, True, True, False. B) True, False, False, True.
C) False, True, False, False. D) True, False, True, True.

92) Which of the following statements is *false*? 92) _____

A) Tuples must store heterogeneous data (that is, data of different types).
B) A tuple's length cannot change during program execution.
C) A tuple's length is its number of elements.
D) Tuples are immutable.

93) Which of the following statements is *false*? 93) _____

A) You can pack a tuple by separating its values with commas.
B) The following statement creates a one-element tuple:

```
a_singleton_tuple = ('red')
```

C) You may surround a tuple's comma-separated list of values with optional parentheses.
D) When you output a tuple, Python always displays its contents in parentheses.

94) Which of the following statements is *false*? 94) _____

A) Assigning a value to a tuple element causes a TypeError.
B) Usually, you iterate over a tuple's elements.
C) Like list indices, tuple indices start at 0.
D) The following snippets create a time_tuple representing an hour, minute and second, display the tuple, then use its elements to calculate the number of seconds since midnight:

```
In [1]: time_tuple = (9, 16, 1)

In [2]: time_tuple
Out[2]: (9, 16, 1)

In [3]: time_tuple[0] * 3600 + time_tuple[1] * 60 + time_tuple[2]
Out[3]: 33361
```

95) Which of the following statements is *false*? 95) _____

A) The += augmented assignment statement can be used with strings and tuples, even though they're immutable.
B) In the following snippets, after the two assignments, tuple1 and tuple2 are two different copies of the same tuple object:

```
In [1]: tuple1 = (10, 20, 30)

In [2]: tuple2 = tuple1

In [3]: tuple2
Out[3]: (10, 20, 30)
```

C) For a string or tuple, the item to the right of += must be a string or tuple, respectively–mixing types causes a TypeError.
D) Concatenating the tuple (40, 50) to tuple1 from Part (b), as in

```
tuple1 += (40, 50)
```

creates a *new* tuple, then assigns a reference to it to the variable tuple1–tuple2 still refers to the original tuple.

96) Which of the following statements a), b) or c) is *false*? 96) _____

A) The following code creates a student_tuple with a first name, last name and list of grades:

```
student_tuple = ('Amanda', 'Blue', [98, 75, 87])
```

B) In the expression student_tuple[2][1], Python views student_tuple[2] as the element of the tuple containing the list [98,75,87], then uses [1] to access the list element containing 75.

C) Even though the tuple in Part (a) is immutable, its list element is mutable.

D) All of the above statements are *true*.

97) Which of the following statements a), b) or c) is *false*? 97) _____

A) You can unpack any sequence's elements by assigning the sequence to a comma-separated list of variables.

B) The following code unpacks a sequence produced by range:

```
number1, number2, number3 = range(3)
```

C) A ValueError occurs if the number of variables to the left of the assignment symbol is not identical to the number of elements in the sequence on the right.

D) All of the above statements are *true*.

98) Which of the following statements a), b) or c) is *false*? 98) _____

A) The following code uses the built-in function list to create a list of tuples containing enumerate's results:

```
colors = ['red', 'orange', 'yellow']
colors_list = list(enumerate(colors))
```

B) The preferred mechanism for accessing an element's index *and* value is the built-in function enumerate, which receives an iterable and creates an iterator that, for each element, returns a tuple containing the element's index and value.

C) The following for loop unpacks each tuple returned by enumerate into the variables index and value and displays them:

```
for index, value in enumerate(colors):
    print(f'{index}: {value}')
```

D) All of the above statements are *true*.

99) Which of the following statements is *false*? 99) _____

A) When taking a slice of a list, the original list is not modified.

B) The following code creates a slice consisting of the elements at indices 2 through 6 of the list numbers:

```
numbers = [2, 3, 5, 7, 11, 13, 17, 19]
numbers2 = numbers[2:6]
```

C) You can slice sequences to create new sequences of the same type containing subsets of the original elements.

D) Slice operations can modify mutable sequences. Slice operations that do *not* modify a sequence work identically for lists, tuples and strings.

100) Which of the following statements is *false*? 100) _____

A) Omitting both the start and end indices on a slice copies the entire sequence.

B) When you specify a slice and omit the starting index, 0 is assumed. So, the slice numbers[:6] is equivalent to the slice numbers[0:6]. If you omit the ending index, Python assumes the sequence's length.

C) With slices, the new sequence's elements refer to the same objects as the original sequence's elements, rather than to separate copies.

D) Slices make deep copies of the sequence's elements.

101) Which of the following statements a), b) or c) is *false*? 101) _____

A) The following code creates a new list in reverse order:

```
numbers[::-1]
```

B) You can use a negative step to select slices in *reverse* order.

C) You can modify a list by assigning to a slice of it–the rest of the list is unchanged.

D) All of the above statements are *true*.

102) Which of the following statements is *false*? 102) _____

A) The following code deletes only the first three elements of numbers by assigning an *empty* list to the three-element slice:

```
numbers[0:3] = []
```

B) The following code assigns a list's elements to a slice of every other element of numbers:

```
numbers = [2, 3, 5, 7, 11, 13, 17, 19]
numbers[2:2:2] = [100, 100, 100, 100]
```

C) When you assign a new object to a variable the original object that the variable referenced will be garbage collected if no other variables refer to it.

D) The following code deletes all the elements in numbers, leaving the existing list empty:

```
numbers[:] = []
```

103) Which of the following statements is *false*? 103) _____

A) The following deletes the list's first two elements:

```
del numbers[0:2]
```

B) The following code creates a list, then uses del to remove its last element:

```
numbers = list(range(0, 10))
del numbers[1]
```

C) The following uses a step in the slice to delete every other element from the entire list:

```
del numbers[::2]
```

D) You can use del with a list to remove the element at any valid index or the element(s) from any valid slice.

104) Which of the following statements is *false*? 104) _____

A) The following code deletes the variable numbers from the interactive session:

```
del numbers
```

B) After deleting numbers from the interactive session, attempting to display it displays an empty list.

C) The del statement can delete any variable.

D) The following code deletes all of the list's elements:

```
del numbers[:]
```

105) Which of the following statements is *false*? 105) _____

A) The function modify_elements multiplies each element of its list argument by 2:

```
def modify_elements(items):
    """Multiplies all element values in items by 2."""
    for i in range(len(items)):
        items[i] *= 2
```

B) When you pass a tuple to a function, attempting to modify the tuple's immutable elements results in a TypeError.

C) Tuples may contain mutable objects, such as lists, but those objects cannot be modified when a tuple is passed to a function.

D) Part (a)'s function modify_elements' items parameter receives a reference to the original list, so the statement in the loop's suite modifies each element in the original list object.

106) Which of the following statements a), b) or c) is *false*? 106) _____

A) Built-in function sorted returns a new list containing the sorted elements of its argument sequence–the original sequence is unmodified.

B) You can use list method sort as follows to arrange a list's elements in ascending order:

```
numbers = [10, 3, 7, 1, 9, 4, 2, 8, 5, 6]
numbers.sort()
```

C) To sort a list in descending order, call list method sort with the optional keyword argument reverse=False.

D) All of the above statements are *true*.

107) Which of the following statements is *false*? 107) _____

A) Often, you'll want to determine whether a sequence (such as a list, tuple or string) contains a value that matches a particular key value.

B) List method index takes as an argument a search key–the value to locate in the list–then searches through the list from index 1 and returns the index of the *first* element that matches the search key.

C) Searching is the process of locating a key in a sequence.

D) List method index raises a ValueError if the value you're searching for is not in the list.

108) Which of the following statements a), b) or c) is *false*? 108) _____

A) You can use *= to *multiply a sequence*–that is, append a sequence to itself multiple times.

B) The following code searches the list numbers for the value 5 starting from index 7 and continuing through the end of the list:

```
numbers.index(5, 7)
```

C) After the following snippet, the list numbers contains two copies of the original list's contents:

```
numbers *= 2
```

D) All of the above statements are *true*.

109) Which of the following statements a), b) or c) is *false*? 109) _____

A) The following looks for the value 7 in the range of elements with indices 0 through 4:

```
numbers.index(7, 0, 4)
```

B) In a call to list method index, specifying the starting and ending indices (as the second and third arguments) causes index to search from the starting index up to but not including the ending index location.

C) The call:

```
numbers.index(5, 7)
```

is equivalent to:

```
numbers.index(5, 7, len(numbers))
```

D) All of the above statements are *true*.

110) Which of the following statements is *false*? 110) _____

A) The built-in function all returns True if all items in its iterable argument are True.

B) Functions any and all are examples of external iteration in functional-style programming.

C) Non-empty iterable objects also evaluate to True, whereas any empty iterable evaluates to False.

D) The built-in function any returns True if any item in its iterable argument is True.

111) Consider the list color_names: 111) _____

```
color_names = ['orange', 'yellow', 'green']
```

Which of the following statements a), b) or c) is *false*?

A) Method insert adds a new item at a specified index. The following inserts 'red' at index 0:

```
color_names.insert(0, 'red')
```

B) Lists also have methods that add and remove elements.

C) You can add a new item to the end of a list with method append.

D) All of the above statements are *true*.

112) Which of the following statements is *false*? 112) _____

A) List method remove deletes the first element with a specified value–a NameError occurs if remove's argument is not in the list.
B) List method copy returns a new list containing a shallow copy of the original list.
C) List method reverse reverses the contents of a list in place.
D) List method count searches for its argument in a list and returns the number of times it is found.

113) Which of the following statements is *false*? 113) _____

A) Python does not have a built-in stack type, but you can think of a stack as a constrained list.
B) You can run out of memory if you keep pushing items faster than you pop them.
C) You pop using list method pop with no arguments, which removes and returns the item at the front of the list.
D) You push using list method append, which adds a new element to the end of the list.

114) Which of the following statements a), b) or c) is *false*? 114) _____

A) List comprehensions provide a concise and convenient notation for creating new lists.
B) List comprehensions can replace many for statements that iterate over existing sequences and create new lists, such as:

```
list1 = []

for item in range(1, 6):
    list1.append(item)
```

We can accomplish the same task in a single line of code with a list comprehension:

```
list2 = [item for item in range(1, 6)]
```

C) The list comprehension's for clause in Part (b) iterates over the sequence produced by range(1, 6). For each item, the list comprehension evaluates the expression to the left of the for clause and places the expression's value (in this case, the item itself) in the new list.
D) All of the above statements are *true*.

115) Which of the following statements a), b) or c) is *false*? 115) _____

A) Mapping is a common functional-style programming operation that produces a result with more elements than the original data being mapped.
B) The following list comprehension uses the expression item**3 to map each value in a range to the value's cube:

```
list3 = [item ** 3 for item in range(1, 6)]
```

C) A list comprehension's expression can perform tasks, such as calculations, that map elements to new values (possibly of different types).
D) All of the above statements are *true*.

116) Which of the following statements is *false*?. 116) _____

A) Filtering typically produces a list with more elements than the data being filtered.

B) A common functional-style programming operation is filtering elements to select only those that match a condition.

C) To do filtering in a list comprehension, use the if clause.

d The following list comprehension includes in list4 only the even values produced by the for clause:

```
list4 = [item for item in range(1, 11) if item % 2 == 0]
```

117) Which of the following statements a), b) or c) is *false*? 117) _____

A) The generator expression in the following for statement squares and returns only the odd values in numbers:

```
numbers = [10, 3, 7, 1, 9, 4, 2, 8, 5, 6]

for value in (x ** 2 for x in numbers if x % 2 != 0):
    print(value, end='  ')
```

B) A generator expression is similar to a list comprehension, but is enclosed in parentheses (rather than []) and creates an iterable generator object that produces values on demand.

C) A generator expression does not create a list.

D) All of the above statements are *true*.

118) Which of the following statements is *false*? 118) _____

A) Python functions are objects that you can assign to variables, pass to other functions and return from functions.

B) The following code uses built-in function filter to obtain the odd values in numbers:

```
numbers = [10, 3, 7, 1, 9, 4, 2, 8, 5, 6]

def is_odd(x):
    """Returns True only if x is odd."""
    return x % 2 != 0

list(filter(is_odd, numbers))
```

C) Function filter returns an iterator, so filter's results are produced immediately.

D) Function filter's first argument must be a function that receives one argument and returns True if the value should be included in the result.

119) Which of the following statements is *false*? 119) _____

A) A lambda expression is an *anonymous function*–that is, a *function without a name.*

B) A lambda explicitly returns its expression's value.

C) In the following filter call the first argument is the lambda:

```
filter(lambda x: x % 2 != 0, numbers)
```

D) For simple functions that return only a *single expression's value*, you can use a lambda expression to define the function inline where it's needed–typically as it's passed to another function.

120) Which of the following statements is *false*? 120) _____

A) The following code uses built-in function map with a lambda to square each value in numbers:

```
list(map(lambda x: x ** 2, numbers))
```

B) Function map uses eager evaluation.

C) Function map's first argument is a function that receives one value and returns a new value–in Part (a), a lambda that squares its argument. Function map's second argument is an iterable of values to map.

D) The equivalent list comprehension to Part (a) is:

```
[item ** 2 for item in numbers]
```

121) Which of the following statements is *false*? 121) _____

A) You also can create custom reductions using the functools module's reduce function.

B) The built-in functions len, sum, min and max perform reductions.

C) In the worlds of big data and Hadoop, MapReduce programming is based on the filter, map and reduce operations in functional-style programming.

D) Reductions process a sequence's elements into a small number of values.

122) Which of the following statements a), b) or c) is *false*? 122) _____

A) Built-in function ord returns the numerical value of a character.

B) Sometimes you'll need to find the minimum and maximum of more complex objects, such as strings.

C) Consider the following comparison:

```
'Red' < 'orange'
```

The letter 'R' "comes after" 'o' in the alphabet, so 'Red' is greater than 'orange' and the condition above is False.

D) All of the above statements are *true*.

123) Which of the following statements a), b) or c) is *false*? 123) _____

A) To identify a particular table element, we specify *two* indices–by convention, the first identifies the element's column, the second the element's row.

B) Lists can contain other lists as elements.

C) Multidimensional lists can have more than two indices.

D) All of the above statements are *true*.

124) Consider a two-dimensional list with three rows and four columns (i.e., a 3-by-4 list) that might represent the grades of three students who each took four exams in a course: 124) _____

```
a = [[77, 68, 86, 73], [96, 87, 89, 81], [70, 90, 86, 81]]
```

Which of the following statements a), b) or c) is *false*?

A) The following nested for statement outputs the rows of the two-dimensional list a one row at a time:

```
for row in a:
    for item in row:
        print(item, end=' ')
    print()
```

B) The element names in the last column all have 3 as the second index.

C) Writing the list as follows makes its row and column tabular structure clearer:

```
a =   [77, 68, 86, 73],  # first student's grades
      96, 87, 89, 81],  # second student's grades
      [70, 90, 86, 81]   # third student's grades
```

D) All of the above statements are *true*.

125) Which of the following statements a), b) or c) is *false*? 125) _____

A) Visualizations help you "get to know" your data.

B) The Matplotlib visualization library is built over the Seaborn visualization library and simplifies many Seaborn operations.

C) Visualizations give you a powerful way to understand data that goes beyond simply looking at raw data.

D) All of the above statements are *true*.

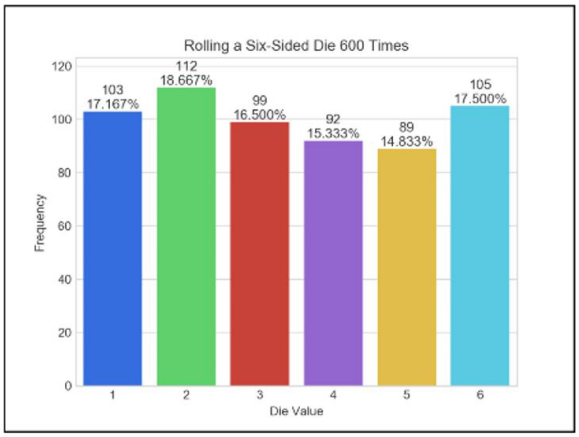
126) Which of the following statements is *false*? 126) _____

A) The larger the number of die rolls, the closer the frequency percentages will be to the expected 16.667%.

B) For 6,000,000 rolls, we expect about 1,000,000 of each face.

C) For 600 die rolls, we expect about 100 occurrences of each die face. As we run a die-rolling simulation for 60,000 die rolls, the bars will become much closer in size. At 6,000,000 die rolls, they'll appear to be exactly the same size. This is the "principal of least privilege" at work.

D) Seaborn refers to the following type of graph as a bar plot:



127) Which of the following statements a), b) or c) is *false*? 127) _____

A) The following code uses a *list comprehension* to create a list of 600 random die values, then uses NumPy's unique function to determine the unique roll values (guaranteed to include all six possible face values) and their frequencies:

```
rolls = [random.randrange(1, 7) for i in range(600)]
values, frequencies = np.unique(rolls, return_counts=True)
```

B) Specifying the keyword argument return_counts=True tells unique to count each unique value's number of occurrences.

C) The NumPy library provides the high-performance ndarray collection, which is typically much faster than lists.

D) All of the above statements are *true*.

128) Which of the following statements is *false*? 128) _____

A) A dictionary is an unordered collection which stores key—value pairs that map immutable keys to values, just as a conventional dictionary maps words to definitions.

B) Lists and tuples are built-in sequence collections.

C) Dictionaries, strings and sets are built-in non-sequence collections.

D) A set is an unordered collection of unique immutable elements.

129) Which of the following statements a), b) or c) is *false*? 129) _____

A) Multiple keys can have the same value, such as two different inventory codes that have the same quantity in stock.
B) A dictionary associates keys with values. Each key maps to a specific value.
C) A dictionary's keys must be mutable (such as strings, numbers or tuples) and unique (that is, no duplicates).
D) All of the above statements are *true*.

130) Which of the following statements is *false*? 130) _____

A) You can create a dictionary by enclosing in curly braces, {}, a comma-separated list of key—value pairs, each of the form *key*: *value*.
B) The following statement creates a dictionary with the country-name keys 'Finland', 'South Africa' and 'Nepal' and their corresponding Internet country code values 'fi', 'za' and 'np':

```
country_codes = {'Finland', 'fi', 'South Africa', 'za',
                 'Nepal', 'np'}
```

C) When you output a dictionary, its comma-separated list of key—value pairs is always enclosed in curly braces.
D) You can create an empty dictionary with {}.

131) Which of the following statements is *false*? 131) _____

A) Method clean deletes the dictionary's key—value pairs:
B) An empty dictionary evaluates to False.
C) You can use a dictionary as a condition to determine if it's empty–a non-empty dictionary evaluates to True.
D) When you pass a dictionary to built-in function len, the function returns the number of key—value pairs in the dictionary.

132) Which of the following statements A), B) and C) is *false*? 132) _____

A) Multiple keys in a dictionary can have the same value.
B) The following for statement iterates through dictionary days_per_month's key—value pairs. Dictionary method items returns each key—value pair as a tuple, which is unpacked into the target variables month and days:

```
for month, days in days_per_month.items():
    print(f'{month} has {days} days')
```

C) The following dictionary maps month-name strings to int values representing the numbers of days in the corresponding month:

```
days_per_month = {'January': 31, 'February': 28, 'March': 31}
```

D) All of the above statements are *true*.

133) Which of the following statements a), b) or c) is *false*? 133) _____

A) The following code creates the dictionary roman_numerals, which maps roman numerals to their integer equivalents (the value for 'X' is intentionally wrong):

```
roman_numerals = {'I': 1, 'II': 2, 'III': 3, 'V': 5, 'X': 100}
```

B) You can update a key's associated value in an assignment statement. The following code replaces the incorrect value associated with the key 'X' in Part (a):

```
roman_numerals['X'] = 10
```

C) The following code gets the value associated with the key 'V' in Part (a):

```
roman_numerals['V']
```

D) All of the above statements are *true*.

134) Which of the following statements a), b) or c) is *false*? 134) _____

A) Assigning a value to a nonexistent key inserts the key—value pair in the dictionary.
B) Assigning a value to a nonexistent key could be a logic error.
C) String keys are case insensitive.
D) All of the above statements are *true*.

135) Which of the following statements is *false*? 135) _____

A) When you iterate over a view, it "sees" a copy of the dictionary's current contents.
B) Dictionary methods items, keys and values each return a view of a dictionary's data.
C) You should not modify a dictionary while iterating through a view. According to the Python Standard Library documentation, either you'll get a RuntimeError or the loop might not process all of the view's values.
D) Dictionary methods keys and values can be used to iterate through a dictionary's keys or values, respectively.

136) Which of the following statements a), b) or c) is *false*? 136) _____

A) An equals (==) comparison evaluates to True if both dictionaries have the same key—value pairs, regardless of the order in which those pairs were added to each dictionary.
B) The comparison operators == and != can be used to determine whether two dictionaries have identical or different contents.
C) In the following dictionary, the keys are lists of three integer student grades and the values are strings representing the students' names.

```
grade_book = {
    'Susan': [92, 85, 100],
    'Eduardo': [83, 95, 79],
    'Azizi': [91, 89, 82],
    'Pantipa': [97, 91, 92]
}
```

D) All of the above statements are *true*.

137) Which of the following statements a), b) or c) is *false*? 137) _____

A) Python automatically concatenates strings separated by whitespace in parentheses.

B) Breaking a string into words is known as tokenizing the string.

C) The following code builds a dictionary to count the number of occurrences of each word in a string named text–the dictionary's keys will be the unique words, and its values will be integer counts of how many times each word appears in text:

```
word_counts = {}

for word in text.split():
    if word in word_counts:
        word_counts[word] += 1
    else:
        word_counts[word] = 1
```

D) All of the above statements are *true*.

138) Which of the following statements about the following code is *false*? 138) _____

```
word_counts = {}

for word in text.split():
    if word in word_counts:
        word_counts[word] += 1
    else:
        word_counts[word] = 1
```

A) Method split returns a list of tokens (that is, the words in text).

B) The expression text.split()tokenizes text by calling string method split, which separates the words using the method's delimiter string argument–if you do not provide an argument, split uses a space.

C) The expression word_counts[word] += 1 inserts a new key-value pair in the dictionary.

D) All of the above statements are *true*.

139) Which of the following statements a), b) or c) is *false*? 139) _____

A) The snippet in Part (b) provided an incorrect country code ('ar') for Australia. The following code corrects this by using a keyword argument to update the value associated with

```
'Australia':
    country_codes.update(Australia='au')
```

B) Method update can convert keyword arguments into key—value pairs to insert. The following call converts the parameter name Australia into the string key 'Australia' and associates the incorrect value 'ar' with that key:

```
country_codes.update(Australia='ar')
```

C) The following call to dictionary method update receives a dictionary containing one key—value pair to insert or update:

```
country_codes.update({'South Africa': 'za'})
```

D) All of the above statements are *true*.

140) Which of the following statements a), b) or c) is *false*? 140) _____

A) Dictionary comprehensions provide a convenient notation for quickly generating dictionaries, often by mapping one dictionary to another.

B) The following code, which has a dictionary with unique values, reverses the key—value pairs, so the values become the keys and vice versa:

```
months = {'January': 1, 'February': 2, 'March': 3}
months2 = {number: name for name, number in months.items()}
```

C) If the values in Part (b) were not unique, then the code would generate a ValueError.

D) All of the above statements are *true*.

141) Which of the following statements is *false*? 141) _____

A) Dictionaries do not support slicing.

B) Sets may contain only immutable objects, like strings, ints, floats and tuples that contain only immutable elements.

C) Sets are iterable, so they are sequences and they support indexing and slicing with square brackets, [].

D) A set is an unordered collection of unique values.

142) Which of the following statements a), b) or c) is *false*? 142) _____

A) The following code creates a set of strings named colors:

```
colors = {'red', 'orange', 'yellow', 'green', 'red', 'blue'}
```

B) Sets are ordered, so you can write code that depends on the order of their elements.

C) Duplicate elimination is automatic when creating a set.

D) All of the above statements are *true*.

143) Which of the following statements is *false*? 143) _____

A) You can check whether a set contains a particular value using the in and not in operators.

B) You can determine the number of items in a set with the built-in len function.

C) Sets are not iterable, so you cannot process each set element with a for loop.

D) Sets are unordered, so there's no significance to the iteration order.

144) Consider the following session: 144) _____

```
In [1]: numbers = list(range(10)) + list(range(5))

In [2]: numbers
Out[2]: [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 0, 1, 2, 3, 4]

In [3]: set(numbers)
Out[3]: ???
```

Which of the following would be displayed where we wrote ??? in Out[3]?

A) {0, 1, 2, 3, 4, 5, 6, 7, 8, 9}

B) {[0, 1, 2, 3, 4, 5, 6, 7, 8, 9]}

C) (0, 1, 2, 3, 4, 5, 6, 7, 8, 9)

D) [0, 1, 2, 3, 4, 5, 6, 7, 8, 9]

145) Which of the following statements a), b) or c) is *false*? 145) _____

A) A frozenset is an immutable set–it cannot be modified after you create it, so a set can contain frozensets as elements.
B) Sets are immutable, so sets can have other sets as elements.
C) The built-in function frozenset creates a frozenset from any iterable.
D) All of the above statements are *true*.

146) Which of the following statements is *false*? 146) _____

A) The argument to issubset or issuperset must be a set.
B) You also can check for an improper subset with the set method issubset.
C) The <= operator tests whether the set to its left is an improper subset of the one to its right–that is, all the elements in the left operand are in the right operand, and the sets might be equal.
D) You can check for an improper superset with the set method issuperset.

147) Which of the following statements is *false*? 147) _____

A) The operands of the binary set operators, like |, must both be any iterable.
B) You can calculate the union with the | operator or with the set type's union method.
C) The set methods may receive any iterable object as an argument.
D) The union of two sets is a set consisting of all the unique elements from both sets.

148) Which of the following statements a), b) or c) is *false*? 148) _____

A) You can calculate the intersection with the & operator or with the set type's intersection method.
B) You can calculate the difference with the - operator or with the a set's difference method.
C) The difference between two sets is a set consisting of the elements in the left operand that are not in the right operand.
D) All of the above statements are *true*.

149) What values are actually displayed where we've inserted ??? in Out[1] and Out[2]: 149) _____

```
In [1]: {1, 3, 5}.isdisjoint({2, 4, 6})
Out[1]: ???

In [2]: {1, 3, 5}.isdisjoint({4, 6, 1})
Out[2]: ???
```

A) False, False B) False, True C) True, True D) True, False

150) Which of the following statements a), b) or c) is *false*? 150) _____

A) Like operator |, union augmented assignment |= performs a set union operation, but |= doesn't modify its left operand.
B) Other mutable set methods are intersection augmented assignment &=, difference augmented assignment -= and symmetric difference augmented assignment ^=. Their corresponding methods with iterable arguments are intersection_update, difference_update and symmetric_difference_update.
C) The set type's update method performs a union operation modifying the set on which it's called–the argument can be any iterable.
D) All of the above statements are *true*.

151) Which of the following statements is *false*? 151) _____

A) Set method add inserts its argument if the argument is *not* already in the set; otherwise, the set remains unchanged.
B) Set method remove removes its argument from a set. A KeyError occurs if the value is not in the set.
C) You can remove the first element of a set with pop.
D) Method discard also removes its argument from a set but does not cause an exception if the value is not in the set.

152) Which of the following statements a), b) or c) is *false*? 152) _____

A) The more die rolls we attempt, the closer each die value's percentage of the total rolls gets to 16.667%. This is a manifestation of Moore's Law.
B) If the random module's randrange function indeed produces integers at random, then every number in the specified range has an equal probability (or likelihood) of being chosen each time the function is called.
C) For six-sided die rolls, each value 1 through 6 should occur about one-sixth of the time, so the probability of any one of these values occurring is $1/6^{\text{th}}$ or about 16.667%.
D) All of the above statements are *true*.

153) Which of the following statements a), b) or c) is *false*? 153) _____

A) For each animation frame, a Matplotlib FuncAnimation calls a function that you define to specify how to change the plot.
B) If we roll only one die per animation frame, we won't be able to run a large number of rolls in a reasonable amount of time. You can increase the execution speed of the simulation by rolling the die more times per animation frame.
C) Displaying animation frames to the screen is relatively fast compared to the die rolls, which occur at the computer's CPU speeds.
D) All of the above statements are *true*.

154) Which of the following statements about FuncAnimation is *false*? 154) _____

A) The FuncAnimation argument fargs (short for "function arguments") is a tuple of other arguments to pass to the function you specify in FuncAnimation's second argument.
B) FuncAnimation has two required arguments–the Figure object in which to display the animation, and a function to call once per animation frame.
C) The FuncAnimation argument interval is the number of milliseconds between animation frames (the default is 200).
D) All of the above statements are *true*.

1) C
2) C
3) B
4) D
5) B
6) A
7) D
8) A
9) B
10) D
11) There are no questions in this section.
12) D
13) C
14) A
15) D
16) D
17) D
18) D
19) C
20) A
21) D
22) B
23) C
24) C
25) A
26) D
27) D
28) D
29) D
30) A
31) B
32) C
33) B
34) C
35) C
36) D
37) A
38) C
39) C
40) A
41) D
42) D
43) B
44) C
45) D
46) B
47) B
48) B
49) A

50) D
51) B
52) B
53) A
54) D
55) D
56) C
57) D
58) D
59) A
60) B
61) B
62) D
63) B
64) D
65) C
66) B
67) B
68) C
69) B
70) B
71) B
72) A
73) C
74) D
75) C
76) B
77) C
78) A
79) D
80) A
81) C
82) B
83) A
84) C
85) A
86) D
87) A
88) D
89) D
90) D
91) D
92) A
93) B
94) B
95) B
96) D
97) D
98) D

99) B
100) D
101) D
102) B
103) B
104) B
105) C
106) C
107) B
108) D
109) A
110) B
111) D
112) A
113) C
114) D
115) A
116) A
117) D
118) C
119) B
120) B
121) D
122) C
123) A
124) C
125) B
126) C
127) A
128) C
129) C
130) B
131) A
132) A
133) D
134) C
135) A
136) C
137) D
138) C
139) D
140) C
141) C
142) B
143) C
144) A
145) B
146) A
147) A

148) D
149) D
150) A
151) C
152) A
153) C
154) D